

Taiwan's Defense Industrial Awakening

The Self-Reliant Defense Strategy and the Porcupine Doctrine

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Executive Summary

Taiwan's defense transformation is one of the most consequential — and underreported — security developments in the Indo-Pacific. The island's \$40 billion supplemental defense budget (approved 2023, disbursed 2023-2027), combined with a fundamental doctrinal shift from conventional military symmetry with China toward asymmetric "porcupine" deterrence, represents a generational rethinking of how a technologically sophisticated democracy defends against a numerically superior neighbour with nuclear weapons and a stated intent to "reunify" Taiwan by force if necessary.

The "porcupine" or Overall Defense Concept (ODC) — developed by former US Admiral Lee Hsi-min while he was Taiwan's Chief of General Staff and championed by former Under Secretary of Defence for Policy David Helvey — redirects Taiwan's defense investment from expensive conventional platforms (submarines, destroyers, advanced fighters) that would be rapidly destroyed in a Chinese assault, toward high-volume, mobile, lethal asymmetric capabilities: coastal anti-ship missile batteries, man-portable anti-aircraft systems, sea mines, loitering munitions, and hardened logistics that deny China a quick victory. The theory of victory is not defeating the PLA — it is making the cost of invasion sufficiently high that Xi Jinping calculates the gamble is not worth taking.

Part I — The Supplemental Budget: \$40 Billion for Deterrence

Scale and Composition

Taiwan's NT\$2.4 trillion (\$40 billion at 2026 exchange rates) supplemental defense budget is the largest special defense appropriation in Taiwan's history. It supplements the regular annual defense budget (approximately NT\$400-420 billion, or approximately \$13-14B per year, representing roughly 2.4% of GDP) and is explicitly focused on the asymmetric deterrence capabilities prioritised under the ODC.

The supplemental budget's allocation prioritises:

- Submarine programme (IDS — Indigenous Defense Submarine): NT\$494 billion (\$16.5B) — the single largest line item, funding construction of 8 domestic submarines (see Part II)
- Hsiung Feng III (HF-3) supersonic anti-ship missiles: 232 additional missiles ordered, bringing total inventory toward operational saturating capability against PLAN surface forces
- Tien Kung III surface-to-air missile system upgrades: Extended range and electronic countermeasure resistance
- Coastal defense cruise missiles (Hsiung Feng II/III land-based): Additional mobile launcher batteries for distributed coastal defense

- Loitering munitions and drone systems: Accelerated domestic development and import programme for suicide drones analogous to Ukraine's Lancet/Shahed deployments
 - Ammunition stockpiling: Significant investment in artillery ammunition and missile reserves — a direct Ukraine lesson applied to Taiwan's situation
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Part II — The Indigenous Defense Submarine Programme

The Most Complex Domestic Defense Programme in Taiwan's History

Taiwan's Indigenous Defense Submarine (IDS) programme, centred on the lead boat "Hai Kun" (launched September 28, 2023, at Kaohsiung), represents Taiwan's most ambitious domestic defense industrial undertaking since the F-CK-1 Ching-kuo Indigenous Defense Fighter of the early 1990s. The Hai Kun displaces approximately 2,500 tonnes submerged, uses diesel-electric propulsion with air-independent propulsion (AIP) capability, and is equipped with Mk-48 torpedoes (sourced from the US) and potentially indigenous Hsiung Feng III anti-ship missiles in a submarine-launched variant.

The programme's significance extends beyond the hardware. Taiwan's domestic submarine construction — undertaken after the US, France, and Germany all declined to directly supply submarine technology transfer due to Chinese diplomatic pressure — required Taiwan to source submarine components from approximately 12-14 nations through commercial channels, a technical intelligence achievement of considerable complexity. The propulsion systems, combat management system, and sonar suite were assembled from European, American, and indigenous components.

Eight IDS submarines are planned over the supplemental budget period (2023-2027 delivery schedule). If successfully completed, Taiwan's submarine force (currently 4 aging boats — 2 ex-US 1970s vintage Tench class and 2 Dutch-built Hai Lung class from the 1980s) would grow to 10 submarines — a force capable of meaningful anti-access operations against PLAN forces in Taiwan's surrounding waters. The tactical value: a credible submarine force complicates PLAN amphibious planning, forces PLAN to allocate ASW (anti-submarine warfare) resources that would otherwise be dedicated to Taiwan's coastal defense, and threatens PLAN logistical lines in a sustained conflict.

Part III — Missiles: The Porcupine's Quills

Hsiung Feng III: Taiwan's Anti-Ship Deterrent

The Hsiung Feng III (HF-3) is Taiwan's supersonic, ramjet-powered anti-ship cruise missile — the core kinetic deterrent against PLAN surface forces. The HF-3 has a reported range of approximately 400km (classified, range estimates vary) and a cruising speed of approximately Mach 2.0, with terminal sprint velocity reported as Mach 2.5-3.0. A saturating HF-3 strike against PLAN amphibious landing ships — delivered from mobile coastal launcher batteries, surface combatants, and potentially submarine tubes — is the scenario most feared by PLAN amphibious planners.

Taiwan's NCSIST (the government defense R&D agency) has been accelerating HF-3 production and developing extended-range variants. The 232 additional HF-3 missiles funded in the supplemental budget represent a significant addition to a pre-supplemental inventory estimated at 100-150 missiles. At full supplemental budget delivery, Taiwan's HF-3 inventory could approach 400+ missiles — a number sufficient to conduct multiple saturating strike salvos against an invasion fleet.

The Yunfeng (Cloud Peak) missile — Taiwan's long-range land-attack cruise missile (LACM), capable of reaching targets in mainland China — has been in service since approximately 2015 in classified numbers. The Yunfeng provides Taiwan with a retaliatory strike capability: the ability to strike Chinese military infrastructure (PLAN ports, PLAABF airbases, PLARF missile launch facilities) in the event of a

cross-strait conflict. This retaliatory capability is Taiwan's analog to the "counterstrike" capability Japan has been developing with Tomahawk missiles — both represent the abandonment of the purely defensive military posture that characterised these democracies for decades.

Part IV — Doctrinal Transformation: From Symmetry to Asymmetry

The ODC (Overall Defense Concept) Implementation

The ODC's core insight is that Taiwan cannot match China's military buildup quantitatively. China's defense budget is approximately 25× Taiwan's regular defense budget; its active-duty military (2.0 million personnel) is approximately 15× Taiwan's (130,000 active). Any symmetric force-building competition would be futile — China simply outbuilds Taiwan faster than Taiwan can respond.

The ODC instead concentrates Taiwan's defense investment in capabilities that are:

1. Affordable at scale: Coastal missiles, sea mines, loitering munitions, and anti-tank systems cost far less per unit than destroyers or fighter aircraft
2. Hard to destroy in a first strike: Mobile, dispersed capabilities are far more survivable than concentrated airbase or naval base targets
3. Effective against the specific threat: PLAN's primary invasion vector is amphibious assault — a large number of landing ships, transports, and escort vessels that must transit the Taiwan Strait and land on Taiwan's beaches. Anti-ship missiles, sea mines, and coastal defense systems are precisely optimised against this threat

The practical implementation of the ODC in the supplemental budget is visible in the procurement priorities: missiles (hundreds, not dozens), mines (sea mine production acceleration), unmanned systems (drones), and hardened logistics (fuel, ammunition, command-and-control survivability). The F-16V fleet upgrade and IDS submarine programme are partial exceptions — conventional platforms justified by their asymmetric contribution (F-16Vs for air defence and ground attack; submarines for sea denial).

The Ukraine Lessons Applied

Russia's invasion of Ukraine in February 2022 provided Taiwan's defense planners with the most current and comprehensive real-world data set for how a democratic nation defends against a larger neighbour's military assault. The lessons Taiwan has drawn — extensively documented in MND public statements and think tank analyses — include:

1. Anti-tank missiles destroy armour: Javelin, NLAW, and Stugna-P performance against Russian tanks confirmed that man-portable anti-tank guided missiles (ATGMs) are effective at scale against an adversary without adequate combined-arms infantry-armour integration
 2. Air defense denies air superiority: Ukraine's MANPADS (Stinger) and Buk/S-300 operations demonstrated that determined, distributed air defense can deny an attacking force the air superiority it needs to enable ground operations
 3. Ammunition stockpiles matter: Ukraine's early ammunition shortage (particularly 155mm artillery shells) nearly lost the war before Western resupply; Taiwan has prioritised ammunition stockpile expansion
 4. Loitering munitions are decisive: Ukraine's Bayraktar TB2 and subsequent FPV drone operations demonstrated asymmetric lethality that Taiwan is explicitly replicating through domestic drone development
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Part V — US Arms Sales and Alliance Architecture

The Informal Alliance and Formal Commitments

Taiwan's security relationship with the United States is governed by the Taiwan Relations Act (1979) — which commits the US to providing Taiwan with sufficient defensive arms and to maintain US capacity to resist "any resort to force" against Taiwan — rather than the formal mutual defense treaty that governs US commitments to Japan, South Korea, and NATO members. This "strategic ambiguity" — the US will not commit in advance to defending Taiwan militarily — is itself a deterrent: China cannot know with certainty whether US military intervention would follow a Taiwan conflict, and therefore cannot plan around US inaction.

Recent US arms sales to Taiwan have been substantial:

- F-16V Viper upgrade: 66 new F-16Vs (delivered from 2023) and upgrade of existing fleet to Block 70/72 standard
 - M1A2T Abrams tanks: 108 tanks (delivery ongoing 2025-2026)
 - Patriot PAC-3 MSE upgrade: Enhanced range and intercept capability
 - Harpoon coastal defense system: 100 Harpoon anti-ship missiles in mobile coastal launcher configuration (delivered 2022-2024)
 - ATACMS surface-to-surface missile system: Under discussion for Taiwan application (would provide significant land-attack range)
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Conclusion: The Deterrent Architecture Taking Shape

Taiwan's defense transformation is real and consequential. The \$40 billion supplemental budget, the IDS submarine programme, the HF-3 missile expansion, and the doctrinal shift to ODC/asymmetric deterrence represent the most serious Taiwan defense modernisation in a generation. Whether it is sufficient to deter Xi Jinping depends on a calculation that Taiwan cannot fully control — how Xi weighs the costs of invasion against the political imperatives of "reunification" within his tenure.

What Taiwan can control is making the cost calculation as unfavourable to China as possible. The porcupine strategy does exactly that: 400+ HF-3 missiles, 10 submarines (if the IDS programme succeeds), distributed coastal defenses, and an American-armed air force create a layered asymmetric deterrent that makes a successful amphibious operation costly, slow, and uncertain. The 2027 window (centenary of the PLA, by which some analysts believe Xi has set an internal target for Taiwan unification) approaches. Whether Taiwan's ongoing defense transformation will be sufficient to deter China is the most important security question in Asia.

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All data from Taiwan MND, CSIS, RAND, SIPRI, DSCA, and cited news sources.